



PHYSICAL AND STRUCTURAL GEOLOGY

GS 21/L

Igneous Rocks



Igneous Rocks

Igneous rocks (*ignis* = *fire*) form as molten rock cools and solidifies. Considerable evidence supports the idea that the parent material for igneous rocks, called *magma*, is formed by partial melting that occurs at various levels within Earth's crust and upper mantle to depths of about 250 kilometers (about 150 miles).

Magma ----- lava ----- crystallization ----- igneous rock



Igneous Rocks

Once formed, a magma body buoyantly rises toward the surface because it is less dense than the surrounding rocks. (When rock melts, it takes up more space and, hence, it becomes less dense than the surrounding solid rock.) Occasionally, molten rock reaches Earth's surface, where it is called **lava**.

Magma is completely or partly molten rock, which when cooled solidifies to form igneous rocks composed of silicate minerals. Most magmas consist of three materials: a liquid component, a solid component, and a gaseous component.

The liquid portion, called **melt**, is composed mainly of mobile ions of the eight most common elements found in Earth's crust—silicon and oxygen, along with lesser amounts of aluminum, potassium, calcium, sodium, iron, and magnesium.

The solid components (if any) in magma are crystals of silicate minerals. As a magma body cools, the size and number of crystals increase.

Igneous Rocks

The gaseous components of magma, called *volatiles*, are materials that vaporize (form a gas) at surface pressures. The most common volatiles found in magma are water vapor (H₂O), carbon dioxide (CO₂), and sulfur dioxide (SO₂), which are confined by the immense pressure exerted by the overlying rocks.

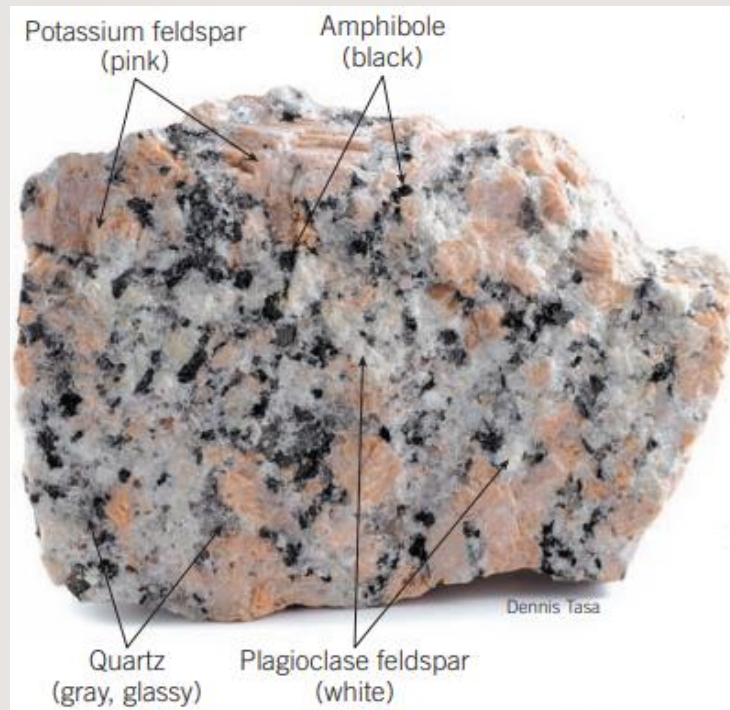


Figure 6.1 Igneous rock composed of interlocking crystals

The rock The largest crystals are about 1 centimeter in length.

Igneous Rocks

From Magma to Crystalline Rock

To better understand how magma crystallizes, let us consider how a simple crystalline solid melts. Recall that, in any crystalline solid, the ions are arranged in a closely packed regular pattern. However, they are not without some motion; they exhibit a restricted vibration about fixed points. As temperature rises, ions vibrate more rapidly and consequently collide with ever-increasing vigor with their neighbors.

In the process called crystallization, cooling reverses the events of melting. As the temperature of the liquid drops, ions pack more closely together as their rate of movement slows. When they are cooled sufficiently, the forces of the chemical bonds will again confine the ions to an orderly crystalline arrangement.

The earliest formed minerals have space to grow and tend to have better-developed crystal faces than do the ones that form later and occupy the remaining spaces. Eventually all of the melt is transformed into a solid mass of interlocking silicate minerals that we call an **igneous rock**.

Igneous Rocks

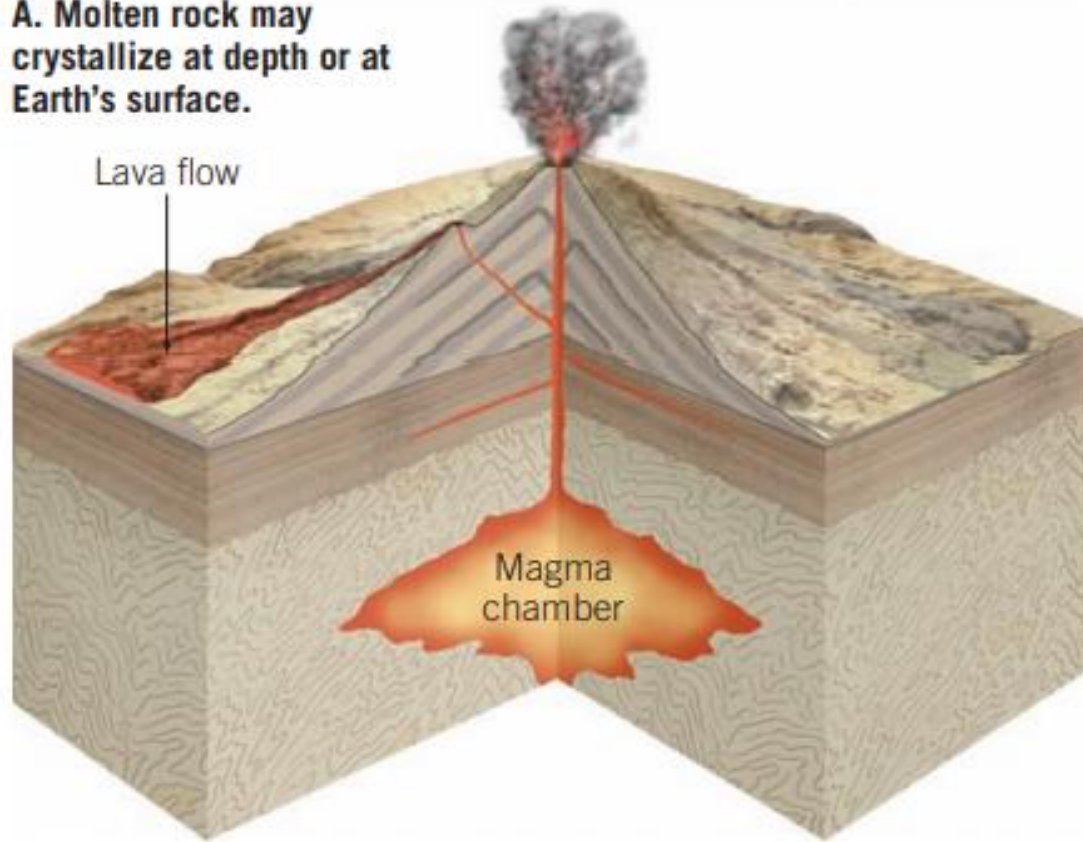
Igneous Processes

Igneous rocks form in two basic settings. Molten rock may crystallize at depth, or it may solidify at Earth's surface. When magma crystallizes at depth, it forms **intrusive igneous rocks**, also known as **plutonic rocks**—after Pluto, the god of the lower world in classical mythology.

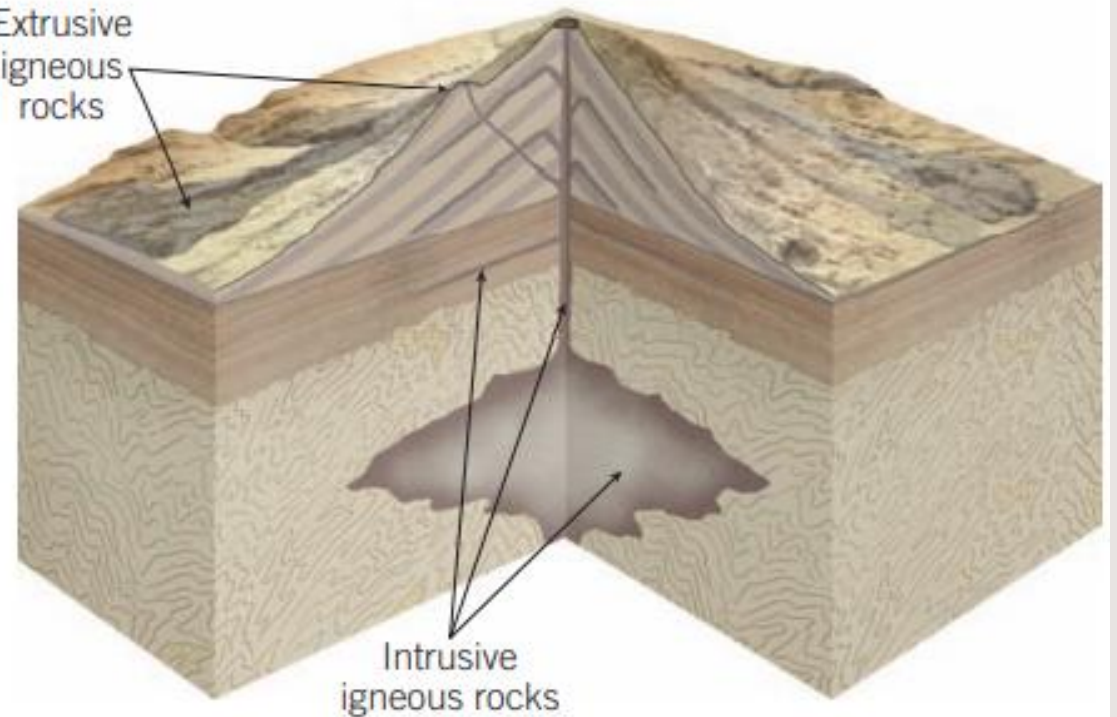
Igneous rocks that form when molten rock solidifies at the surface are classified as **extrusive igneous rocks**. They are also called **volcanic rocks**—after Vulcan, the Roman fire god. Extrusive igneous rocks form when lava solidifies or when volcanic debris falls to Earth's surface. Extrusive igneous rocks are abundant in western portions of the Americas, where they make up the volcanic peaks of the Cascade Range and the Andes Mountains. In addition, many oceanic islands, including the Hawaiian chain and Alaska's Aleutian Islands, are composed almost entirely of extrusive igneous rocks.

Igneous Rocks

A. Molten rock may crystallize at depth or at Earth's surface.



Extrusive
igneous
rocks



B. When magma crystallizes at depth intrusive igneous rocks form. When it solidifies on Earth's surface extrusive igneous rocks form.

Figure 6.2 Intrusive versus extrusive igneous rocks

Igneous Rocks

Igneous rocks grouped according to composition

1. **Granitic (felsic)** - composed mainly of non ferromagnesian minerals such as quartz (~70%) and potassium feldspar. In addition to quartz and feldspar, most granitic rocks contain about 10 percent dark silicate minerals, usually biotite mica and amphibole.
2. **Basaltic (mafic)** - Rocks that contain at least 45 percent dark silicate minerals and calcium-rich plagioclase feldspar.
3. **Andesitic (intermediate)** - Intermediate rocks contain at least 25 percent dark silicate minerals, mainly amphibole, pyroxene, and biotite mica, with the other dominant mineral being plagioclase feldspar

Igneous Rocks

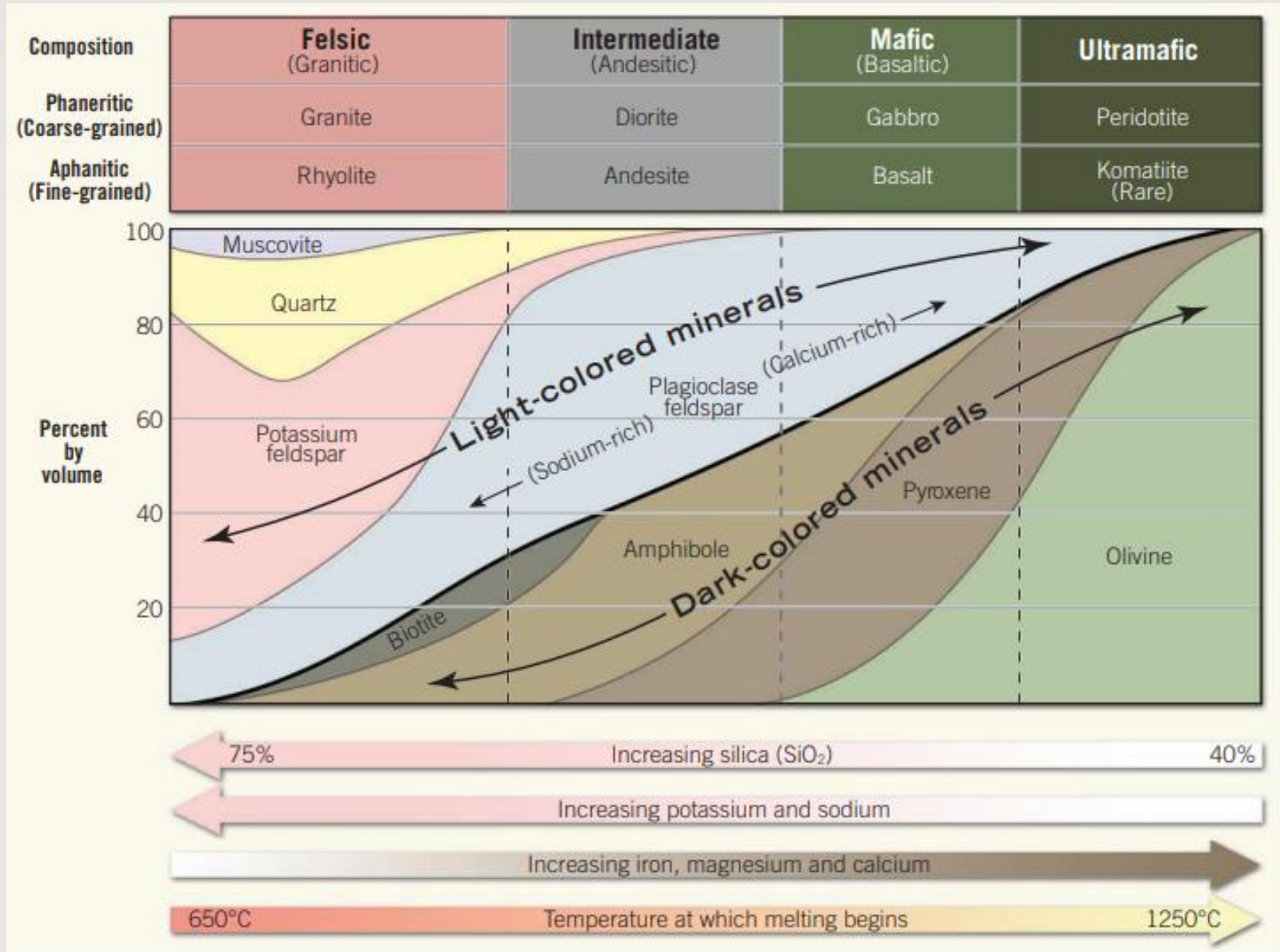


Figure 6.3 Mineralogy of common igneous rocks
(After Dietrich, Daily, and Larsen)

Igneous Rocks

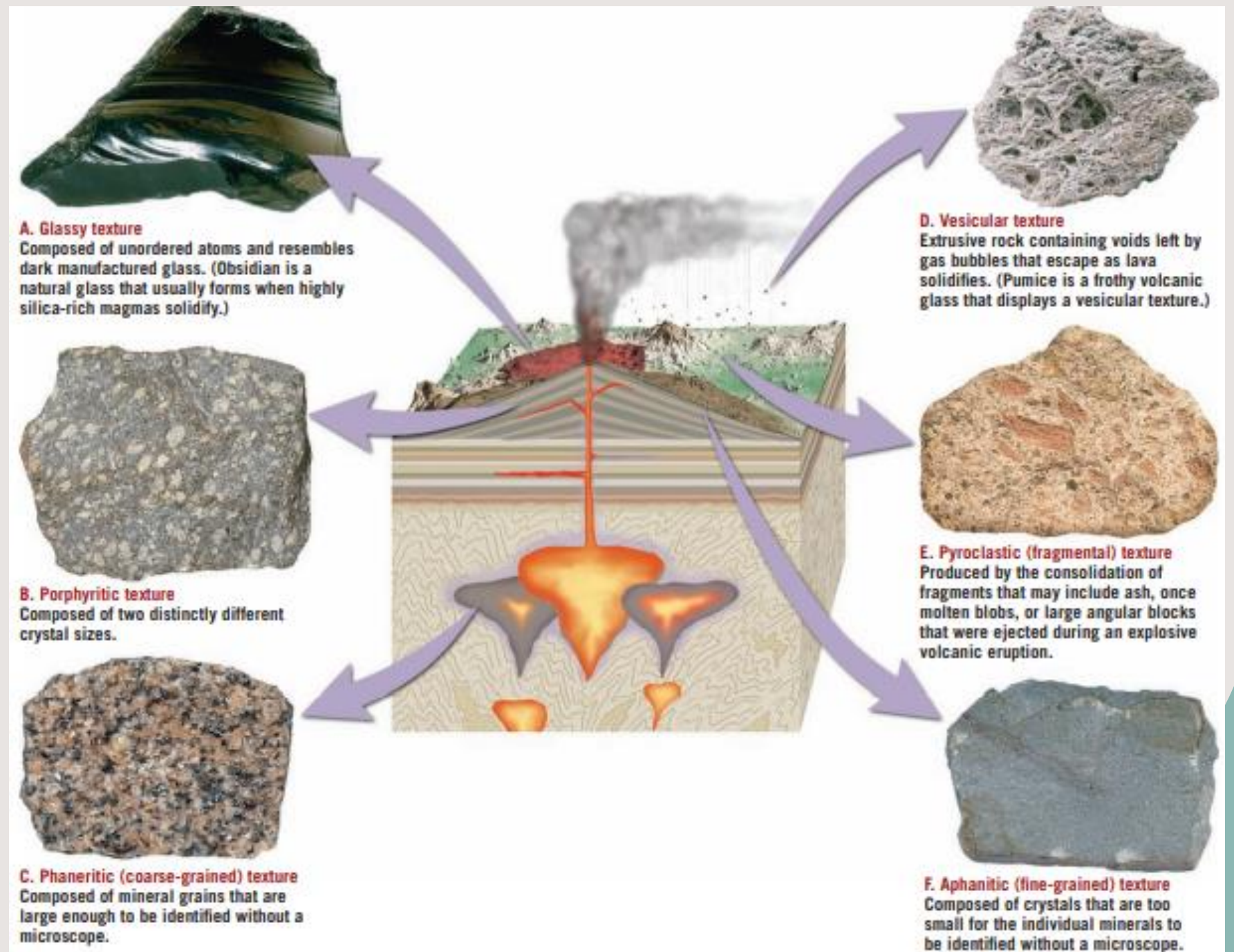
Silica as Composition Indicator

- **High silica rocks** generally are felsic in nature, magmas of such composition are viscous and at least have lower eruption temperatures (~650 C) (low metal content=low melting temperature)
- **Low silica rocks** generally are mafic in nature, with its magmas known to be more fluid because of its high temperature (1050-1250 c). (high metal content=high melting temperature)

Figure 6.4 Classification of igneous rocks
 The classification of igneous rocks is based on mineral composition and texture.
 (Photos by Dennis Tasa and E. J. Tarbuck)

IGNEOUS ROCK CLASSIFICATION CHART					
		MINERAL COMPOSITION			
		Felsic (Granitic)	Intermediate (Andesitic)	Mafic (Basaltic)	Ultramafic
Dominant Minerals		Quartz Potassium feldspar	Amphibole Plagioclase feldspar	Pyroxene Plagioclase feldspar	Olivine Pyroxene
Accessory Minerals		Plagioclase feldspar Amphibole Muscovite Biotite	Pyroxene Biotite	Amphibole Olivine	Plagioclase feldspar
TEXTURE	Phaneritic (coarse-grained)	 Granite	 Diorite	 Gabbro	 Peridotite
	Aphanitic (fine-grained)	 Rhyolite	 Andesite	 Basalt	 Komatiite (rare)
	Porphyritic (two distinct grain sizes)	 Granite porphyry	 Andesite porphyry	 Basalt porphyry	Uncommon
	Glassy	 Obsidian	Less common	Less common	Uncommon
	Vesicular (contains voids)	 Pumice (also glassy)	 Scoria		Uncommon
	Pyroclastic (fragmental)	 Tuff or welded tuff Most fragments < 4mm	→	 Volcanic breccia Most fragments > 4mm	← Uncommon
Rock Color (based on % of dark minerals)		0% to 25%	25% to 45%	45% to 85%	85% to 100%
					

Figure 6.5 Igneous rock textures



Reference

- Tarbuck, Lutgens and Tasa (2014), Earth: An Introduction to Physical Geology 11th ed., Pearson Education, Inc